TOPPERS ACADEMY
SCIENCE & COMMERCE

TOPPERS ACADEMY

Exam Name :-uygr

Date :-16/12/2025

Time :-50 Minutes

Mark :- 50

1. The maximum height attained by a projectile is increased by 10% . Keeping the angle of projection constant, what is percentage increase in the time of flight?

(A) 5% (B) 10% (C) 20% (D) 40%

Ans:-(A)

Solution:-(A)

$$\frac{\mathrm{h}}{\mathrm{T}^{2}}=\frac{\mathrm{u}^{2}\sin^{2}\theta}{2\,\mathrm{g}}\times\frac{\mathrm{g}^{2}}{4\mathrm{u}^{2}\sin^{2}\theta}=\frac{\mathrm{g}}{8}$$

Hence $$\frac{\Delta\mathrm{h}}{\mathrm{h}}=2\frac{\Delta\mathrm{T}}{\mathrm{T}}\text{ i.e. }\frac{\Delta\mathrm{T}}{\mathrm{T}}=\frac{1}{2}\frac{\Delta\mathrm{h}}{\mathrm{h}}$$

$$\text{or }100\times\frac{\Delta\mathrm{T}}{\mathrm{h}}=\frac{1}{2}\left[\frac{\Delta\mathrm{h}}{\mathrm{h}}\times100\right]=\frac{10}{2}=5\%$$

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2. A glass marble projected horizontally from the top of a table falls at a distance x from the edge of the table. If h is the height of the table, then the velocity of projection is

(A) $h\sqrt{\frac{g}{2x}}$ (B) $x\sqrt{\frac{g}{2h}}$ (C) gxh (D) $gx+h$

Ans:-(B)

Solution:-(b) $t=\sqrt{\frac{2h}{g}}, x=v\sqrt{\frac{2h}{g}}$ or $v=x\sqrt{\frac{g}{2h}}$

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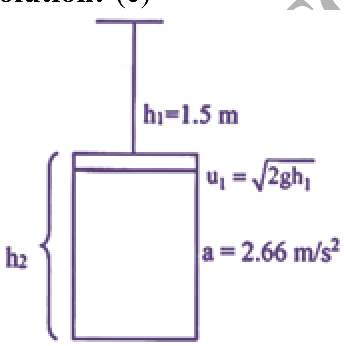
3. A small steel ball is dropped from a height of 1.5 m into a glycerine jar. The ball reaches the bottom of the jar 1.5 second after it was dropped. If the retardation is $2.66\ \mathrm{m/s^2}$, the height of the glycerine in the jar is about (acceleration due to gravity $g=9.8\ \mathrm{m/s^2}$) [MHT-CET 2023]

(A) 7.5 m (B) 7.0 m (C) 5.5 m (D) 3.2 m

Ans:-(C)

Solution:-(c)



Initial height of steel of ball $h_1=1.5$ m

From surface of glycerine Retardation in glycerine $=\mathrm{a}=2.66\ \mathrm{m/s^2}$

Velocity ' u_1 ' at the surface of glycerine

$\sqrt{2gh_1}=\sqrt{2\times9.8\times1.5}$

$\therefore\ u_1=5.4\ \mathrm{m/s}$ Now, ball get at rest at the bottom of surface

$\therefore \mathrm{V}=0\ \mathrm{m/s}$ Now, $v^2=u_1^2-2a_2$

$0=(5.4)^2-2\times2.66\times h_2 \therefore\ \mathrm{h}_2=\frac{29.4}{2\times2.66}\ \therefore\ \mathrm{h}_2=5.5$ m

4. The engine of an aeroplane during take-off exerts a force of 150×10^3 N. Mass of aeroplane is 25×10^3 kg. If the take-off speed is 60 m/s, the length of the runway required is [MHT-CET 2021]
(A) 200 m (B) 100 m (C) 400 m (D) 300 m
Ans:-(D)
Solution:-(d) $F = 150 \times 10^3$ N, $M = 25 \times 10^3$ kg,
$V = 60$ m/s, $u = 0$
Acceleration, $a = \frac{F}{M} = \frac{150\times 10^3}{25\times 10^3} = 6$ m/s^2
$v^2 = u^2 + 2as = 2as \ (\because u = 0)$
$\therefore s = \frac{v^2}{2a} = \frac{60\times 60}{2\times 6} = 300$ m

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5. A stone is projected vertically upwards with speed ' V '. Another stone of same mass is projected at an angle of 60° with the vertical with the same speed ' V '. The ratio of their potential energies at the highest points of their journey is $[\sin 30^\circ = \cos 60^\circ = 0.5, \cos 30^\circ = \sin 60^\circ = \sqrt{3}/2]$ [MHT-CET 2023]
(A) $2:1$ (B) $4:1$ (C) $3:2$ (D) $1:1$
Ans:-(B)
Solution:-(b) When the stone is projected vertically, the maximum height reached is given by $h = \frac{v^2}{2g}$ When it is projected at an angle 30° with horizontal, the maximum height reached is $h' = \frac{v^2 \sin^2 30^\circ}{2g}$ The ratio of the potential energies
$\frac{E_1}{E_2} = \frac{mgh}{mgh'} = \frac{h}{h'} = \frac{1}{\sin^2 30^\circ} = 4$

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6. A car moving at a speed ' V ' is stopped in a certain distance when the breaks produce a deceleration ' a '. If the speed of the car is ' nv ', what must be the decclcration of the car to stop it in the same distance and in the same time? [2025]
(A) n.a (B) $n^2 \cdot a$ (C) $n^3 \cdot a$ (D) $\sqrt{n}, a$
Ans:-(B)
Solution:-(b) $v_f^2 - v_i^2 = 2$ as
For case I $- v_f = 0, v_i = v$
$0^2 - v^2 = 2as \Rightarrow s = \frac{-v^2}{2a}$.(I)
For case II- $v_f = 0, v_i = nv$
$0^2 - n^2v^2 = 2a's \Rightarrow s = \frac{-n^2v^2}{2a'}$.(II)
Car stops with the same distance for both cases, $\frac{-v^2}{2a} = \frac{-n^2v^2}{2a'}$ ... [From (i) and (ii)]
$a' = n^2 a$

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7. A ball is dropped from the tower of height ' h '. The total distance covered by it in last second of its motion is equal to the distance covered by it in first 3 seconds. The value of ' h ' is ($g = 10$ ms^{-2}) [MHT-CET 2022]
(A) 80 m (B) 100 m (C) 125 m (D) 200 m
Ans:-(C)
Solution:-(c) Distance covered in first 3 seconds is given by
$y = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (3)^2 = 45$ m